

John Alexandre Borkowski

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EDUCATION

The University of Texas at Austin, Austin, TX

September 2026

Bachelor of Science in Computer Science and Mathematics, Minor in Business | GPA: 3.55

Relevant Coursework: *Algorithms & Complexity, Data Structures, Computer Architecture, Probability, Statistics, Operating Systems, Machine Learning, Artificial Intelligence, Linear Algebra, Finance, Quantum Informatics*

SKILLS

Languages: Python, C++, C, Java, SQL, R, JavaScript, TypeScript

Libraries & Frameworks: Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, PyTorch, TensorFlow, React, Flask, Django

Infrastructure & Tools: Linux, Bash, AWS (S3, boto3), GCP, Docker, Git, MongoDB

Spoken Languages: Fluent in English, French, and Spanish

EXPERIENCE & CAMPUS INVOLVEMENT

Goldman Sachs, Sales & Trading Summer Analyst (Quantitative Trading), West Palm Beach, FL June 2026 - August 2026

- Built hierarchical Bayesian Nelson-Siegel models to fit issuer-level credit curves, using sector curves as a prior to shrink sparsely-issued names and produce stable term structures for issuers with too few bonds to fit standalone.
- Developed factor-decomposition and covariance-estimation components of an FICC credit risk model with another intern, separating issuer excess returns into systematic factor exposures and idiosyncratic residuals.
- Ran P&L attribution and hedge-ratio analysis on a basis trade in HYG vs. CDX.HY alongside two full-time traders, tracking cash-synthetic dislocation and sizing the Treasury leg to isolate spread basis from rates duration.
- Built an S3-backed caching layer with concurrent fetches across the model's pipeline, checkpointing intermediate results between cross-sectional and factor regression stages to eliminate recomputation on backfills.

Alliance Bernstein, Software Engineer Intern, Nashville, TN

June 2025 - August 2025

- Developed and backtested alpha signaling strategies for the Quantitative Portfolio Modeling Team, implementing **Hidden Markov** and **Mixture of Experts** models for regime classification using **Python**, **Pandas**, and **Scikit-learn**.
- Improved predictive accuracy and identified three novel alpha signals through robust model validation.
- Engineered a secure, encrypted application view for iOS and Android to serve sensitive financial documents, independently enhancing portfolio accessibility for clients representing over \$100 billion in AUM.

Austin Venture Strategy, Product Analyst, Austin, TX

August 2022 - May 2025

- Collaborated with UT tech startups to build software prototypes and MVPs using **Python**, **React**, **Flask**, and **SQL**, driving a **15% increase in user engagement** across launches.

University of Texas Wakeboarding Team, Treasurer, then *President*, Austin, TX

August 2023 - May 2026

- Built and deployed a web application consolidating membership registration and boat-pull scheduling, replacing manual signup coordination for a 150-member club.

PROJECTS

Quantum Chess Engine

December 2025

- Engineered a custom game engine that integrates quantum mechanics into chess, implementing superposition and entanglement of pieces through manual complex-vector state-space modeling.
- Developed the underlying **linear algebra** library to handle unitary transformations and state collapses, simulating non-deterministic outcomes within a strategic framework.

FinDB

August 2024

- Developed a Django stock trading web application integrating real-time market data retrieval, portfolio management, trade execution, and a volatility screener pricing market-implied vols via Bjerk Sund-Stensland.
- Backend built in Python, Django, and SQL, with a query layer supporting live data analysis at interactive latency.

Pipelined CPU / Cache Simulator with Pintos-based OS

April 2024

- Developed a pipelined CPU and cache simulator in C/Assembly with hazard detection and cache replacement policies; extended a Pintos-based OS with multithreading and virtual memory subsystems.